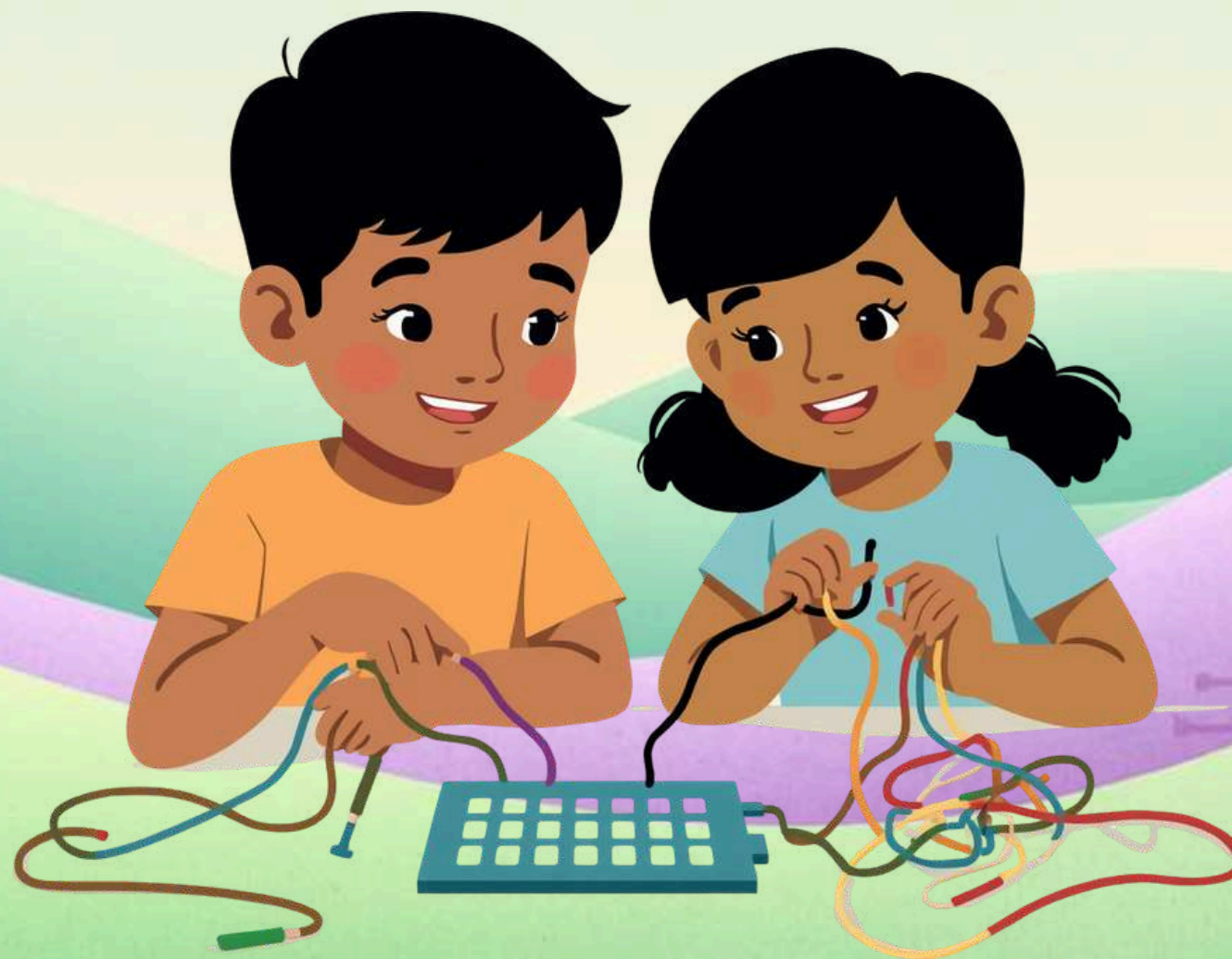


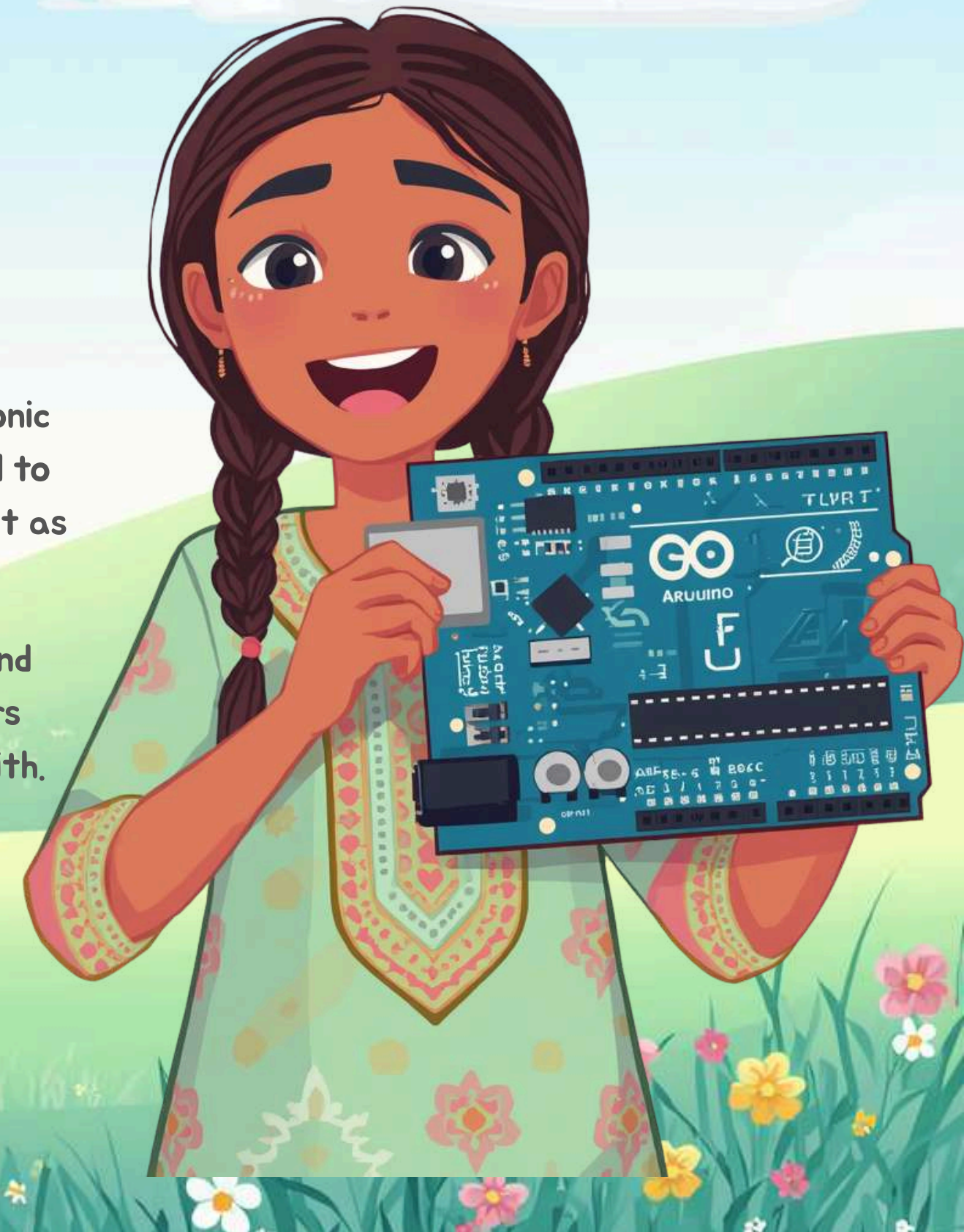
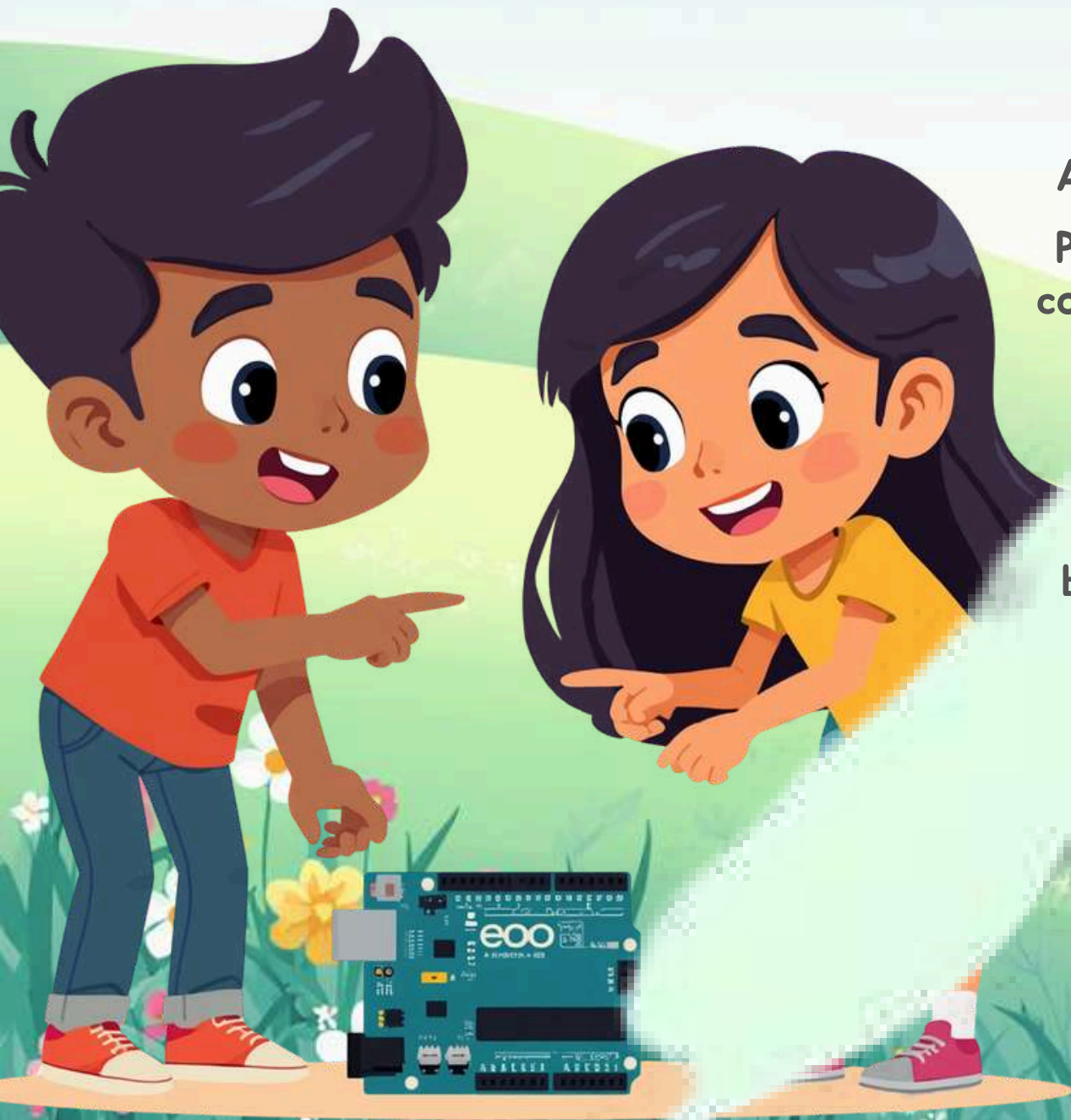
Arduino Fun: Keypad + Password Lock Project



What is Arduino?

Arduino is a small computer that helps us build cool electronic projects! It's a tiny circuit board that can be programmed to control lights, motors, sensors, and so much more. Think of it as the brain of your electronic creations!

With Arduino, you can make things light up, move, beep, and respond to buttons or sensors. It's perfect for beginners because it's easy to learn and super fun to experiment with. Let's explore this amazing tool together!

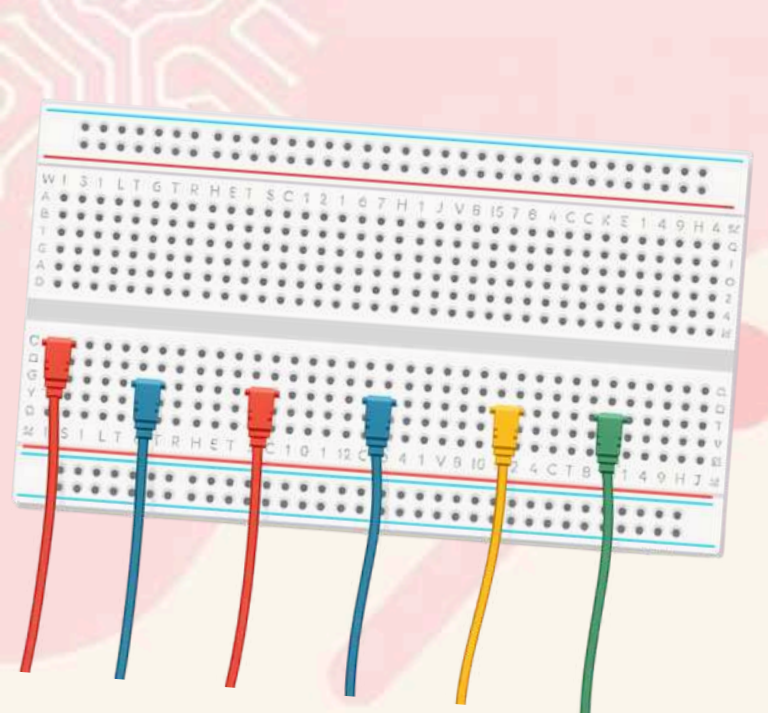


Today's Project Overview

In this exciting project, you will learn to use a keypad to enter a secret password and control a lock mechanism with Arduino. We will explore basic wiring connections and write simple code together!

By the end of this project, you will understand how electronic components work together. You will be able to build your own password-protected lock system – just like the ones used in real security devices!





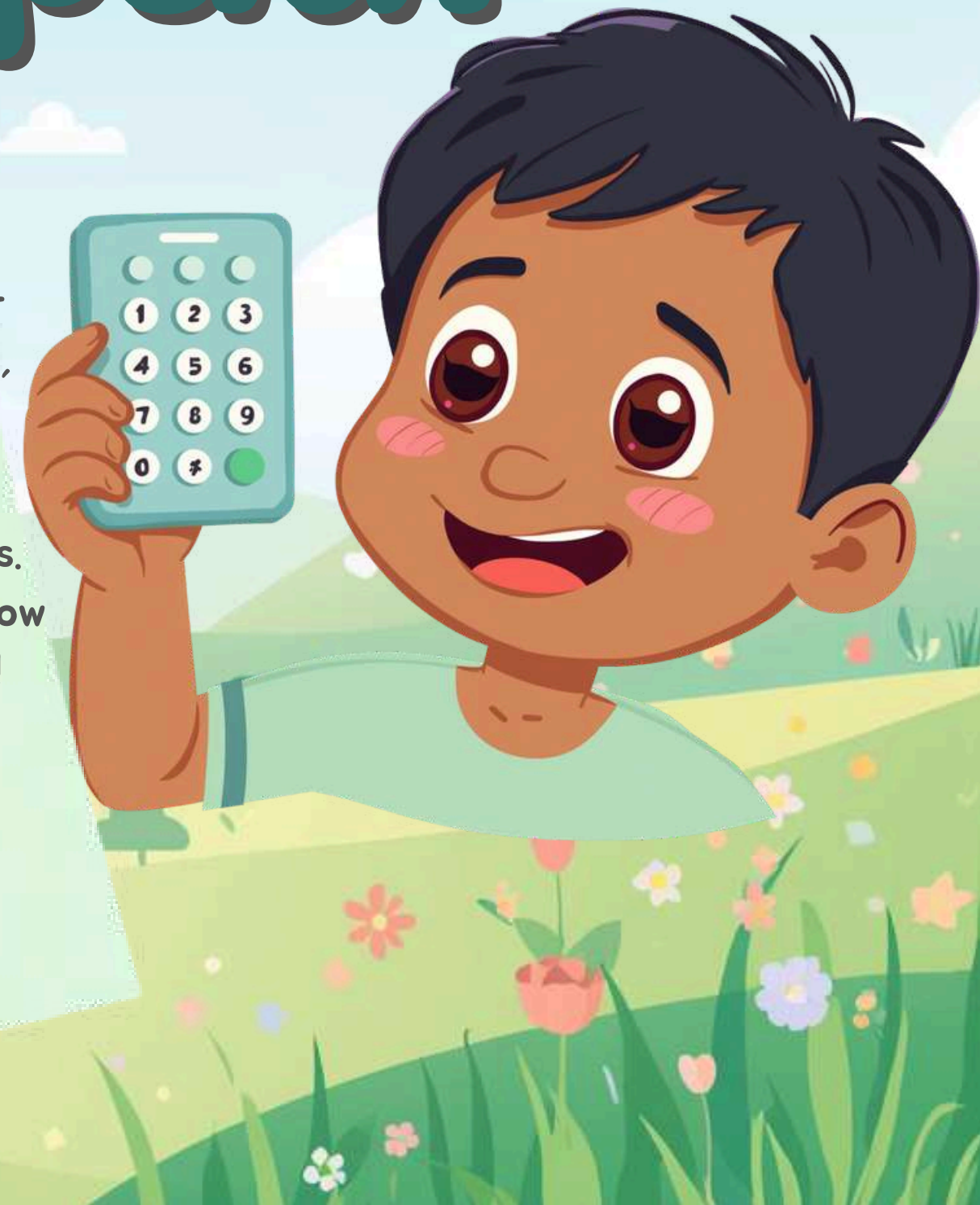
Components Needed



What is a Keypad?

A keypad lets us press numbers to send commands to Arduino. It works like a calculator or phone buttons - when you press a key, it sends a signal to the Arduino telling it which button was pressed!

The 4x4 keypad has 16 buttons arranged in rows and columns. Each button has a unique code that Arduino can read. This is how we can type passwords and control our electronic projects!



Wiring the Keypad to Arduino

The 4x4 keypad has 8 pins - 4 for rows and 4 for columns. Connect the row pins (R1-R4) to Arduino digital pins 9, 8, 7, and 6. Connect the column pins (C1-C4) to digital pins 5, 4, 3, and 2.

Double-check each wire connection before powering on! The keypad sends signals through rows and columns to detect which button you press. This is called a matrix keypad system.

What is a Password Lock?

A password lock opens only when the right code is entered. It keeps things safe by allowing access only to people who know the secret combination!

How does it work? A servo motor acts as the lock mechanism. When you enter the correct password on the keypad, Arduino sends a signal to the servo motor to turn and unlock the door or box!



Coding Basics



We write instructions called code to tell Arduino what to do. Code is like a recipe that Arduino follows step by step. First, we set up the keypad and servo. Then we create a secret password to check against!

Password Checking Logic: When you press keys, Arduino remembers each number. After you enter all digits, it compares your input to the stored password. If they match, the servo unlocks the door. If wrong, it asks you to try again!





Sample Arduino Code Walkthrough

Reading Keypad Input:

```
char key = keypad.getKey();
```

This line checks which button you pressed on the keypad and stores it.

Comparing to Password:

```
if (inputCode == password)
```

This checks if what you typed matches the secret code!

Activating the Servo:

```
servo.write(90);
```

When the password is correct, this line tells the servo motor to turn and unlock!

Wrong Password:

```
servo.write(0);
```

If wrong, the lock stays closed and you can try again.

Testing Your Project

Press the keys to enter your password. The lock will open if correct! Follow these steps to test your Arduino password lock project and see your hard work come to life.

Connect your Arduino to power, then try entering different codes on the keypad. Watch the servo motor - it should move to unlock when you enter the correct password. If wrong, try again!



Troubleshooting Tips

Check all wiring connections carefully - loose wires are the most common problem! Make sure each wire is firmly connected to the correct pin. If your keypad isn't responding, double-check that your code pin numbers match your actual wiring setup.

If your lock doesn't move, test the servo motor separately with a simple code first. Sometimes restarting your Arduino by unplugging and plugging it back in can fix strange behaviour. Don't worry - troubleshooting is part of learning!



Safety Tips

Handle Electronics Gently

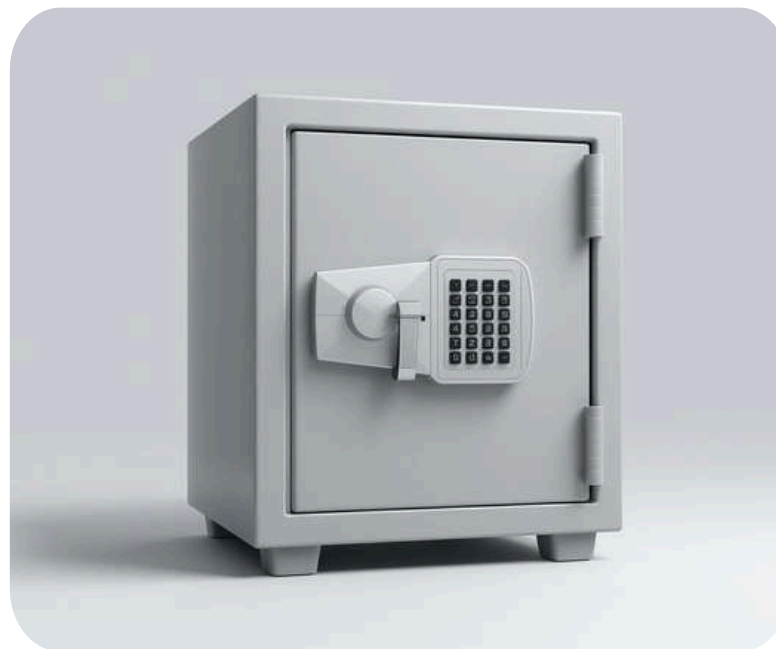
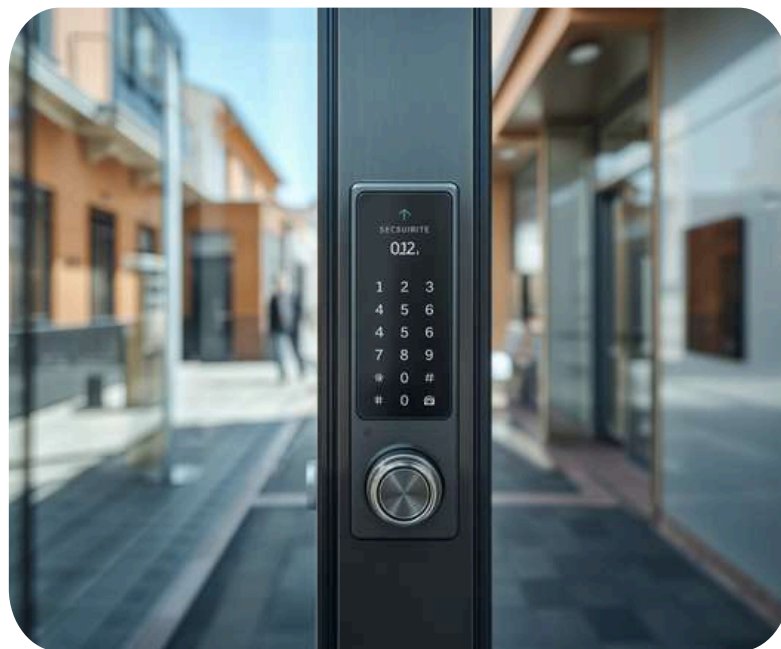
Always pick up and place Arduino and components carefully. Avoid dropping or bending parts. Keep liquids away from your workspace to prevent damage to electronic components.

Ask for Help & Stay Safe

Never power Arduino with wrong voltage - use USB or recommended batteries only. Always ask an adult for help when connecting power. Keep your workspace clean and organized!



Real-World Applications



Summary of What You Learned

Arduino Basics & Components

You learned that Arduino is a small computer that helps build electronic projects. You identified key parts like the Arduino Uno board, keypad, servo motor, breadboard, and jumper wires.

Keypad, Wiring & Coding

You discovered how a 4x4 keypad sends commands to Arduino. You practiced connecting wires and wrote code to check passwords and control the lock mechanism!





Quiz Time!

1. What does Arduino do?
2. How do you enter the password using the keypad?
3. What happens if the password is wrong?

Think about what you learned today!

Discuss your answers with a partner before sharing with the class.

Remember: There are no wrong answers when you're learning!



**Thank You!
Great job! Ready to
build your own
projects?
Keep experimenting
and learning!**